

P5 Planning, analysis and evaluation

Practical · 2026 examination · Concept guide and original exam-style practice

Understand the concept

State a testable relation, specify controls, propose apparatus and a safe repeatable method. Linearise a relationship to extract constants from gradient/intercept, propagate uncertainties and compare predicted values within those uncertainties. Improvements should target the dominant error mechanism.

Worked example: follow the reasoning

If $y=ax^2$, plot y against x^2 : gradient a . A straight line through origin supports the model within measurement uncertainty.

Quick check · multiple choice

A To test $F=kv^2$, what should be plotted against v^2 ? [1]

- A Force F
- B Time t
- C Mass m
- D Velocity v

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A model predicts $F = kv^2$. State graph axes to determine k and describe a check on experimental validity. [3]

Final answer: _____

Answers and explanation

Quick check: A. Force F

Question 1 [3 marks]

Plot F against v^2 [1]; gradient gives k [1]; inspect straight-line fit, intercept and uncertainties/repeats [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.